Lung Nodule Segmentation using Vision Mamba

Uday Yerraballi-EC23B1007

Chris Jason Baskar-CS23B1012

Overview

Lung cancer is one of the leading causes of cancer related deaths worldwide.

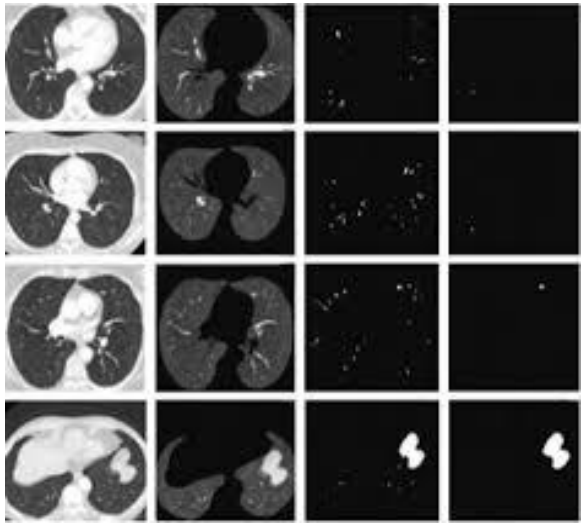
Early detection of lung nodules in CT scans improves survival rates significantly.

- Challenges include:
- Lung nodules are very small and irregular
- Nodules resemble surrounding blood vessels and tissues
- Manual annotation by radiologists is time consuming

Deep learning segmentation models help automatically detect suspicious regions

Dataset information

- LIDC-IDRI (Lung Image Database Consortium)
- Public dataset of thoracic CT scans
- Contains more than 1000 patient scans
- Annotated by four expert radiologists
- Provides labeled lung nodules and segmentation masks
- It is widely used as a benchmark dataset for lung nodule detection
- research.



Citations/Literature Survey

- https://www.sciencedirect.com/science/article/pii/S2667099226000095
- https://arxiv.org/abs/2401.09417

Lung-Mamba: Lung nodule segmentation model optimized by Mamba's selective state spaces

Hadrien T. Gayap, Moulay A. Akhloufi

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Highlights

- Lung-Mamba: A U-Net and Mamba-based model for pulmonary nodule segmentation.
- Integration of Selective State Space (SSM) blocks to capture long-range dependencies.
- Achieves a Dice score of 96.48% on the LIDC-IDRI dataset.

Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model

Lianghui Zhu [1*] Bencheng Liao [2 1*] Qian Zhang [3] Xinlong Wang [4] Wenyu Liu [1] Xinggang Wang [1]

Xiv:2401.09417v3 [cs.CV] 14 Nov 2024

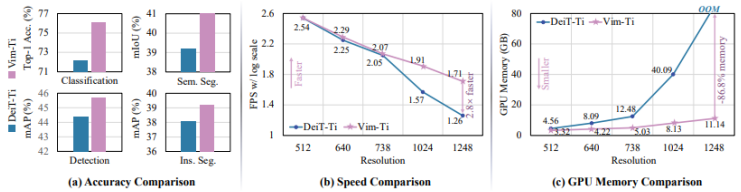


Figure 1: Performance and efficiency comparisons between DeiT (Touvron et al., 2021a) and our Vim model. Results show that Vim outperforms DeiT on both ImageNet classification and downstream detection and segmentation tasks and is more computation and memory efficient than DeiT in dealing with high-resolution images. For example, Vim is 2.8× faster than DeiT and saves 86.8% GPU memory when performing batch inference to extract features on images with a resolution of 1248×1248, *i.e.*, 6084 tokens per image.

Abstract

Recently the state space models (SSMs) with efficient hardware-aware designs, i.e., the Mamba deep learning model, have shown great potential for long sequence modeling. Meanwhile building efficient and generic vision backbones purely upon SSMs is an appealing direction. However, representing visual data is challenging for SSMs due to the position-sensitivity of visual data and the requirement of global context for visual understanding. In this paper, we show that the reliance on self-attention for visual representation learning is not necessary and propose a

semantic segmentation tasks, Vim achieves higher performance compared to well-established vision transformers like DeiT, while also demonstrating significantly improved computation & memory efficiency. For example, Vim is 2.8× faster than DeiT and saves 86.8% GPU memory when performing batch inference to extract features on images with a resolution of 1248×1248. The results demonstrate that Vim is capable of overcoming the computation & memory constraints on performing Transformer-style understanding for high-resolution images and it has great potential to be the next-generation backbone for vision

Architecture

CT scans image is given
and we add light noise
mask to nodule regions

↓

Diffusion Loop.

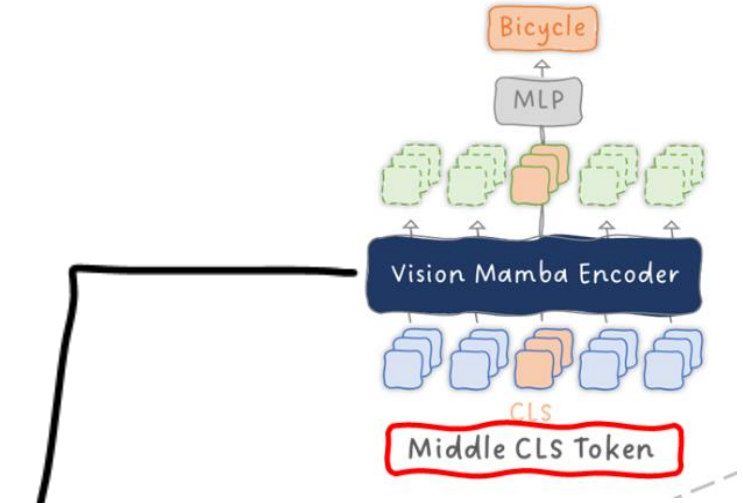
Repeats
step
20 - 50 x

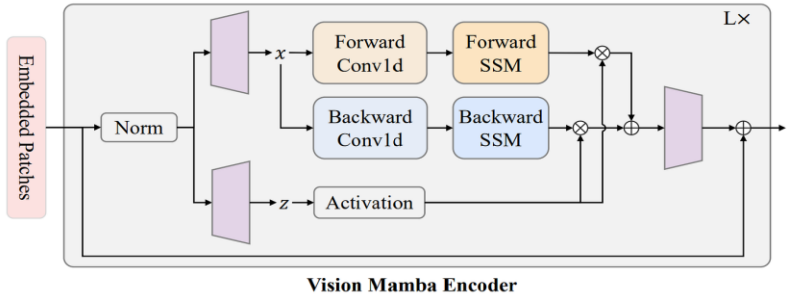
- Feed noisy image to Vision Mamba which patches the image to a sequence.
- scans sequence bidirectionally
- ~~Vis~~ VMamba predicts and removes noise with long range context.

↓

slightly cleaner image is obtained each step, on repetition of diffusion loop we get a noise free nodule mask

HONOR X7b
50MP Dual Camera





Vision Mamba Encoder

Why do we use Vision Mamba for lung nodule segmentation?

- Said to be one of the best alternatives to transformers in image tasks (segmentation,object detection and classification)
- Much more memory efficient with respect to VIT as it memory complexity is linear as opposed to transformers which have an quadratic memory complexity.
- Beats Deit by MetaAI on popular datasets such as CoCo object detection and ImageNet classification.
- Uses a bidrectional state space model which gives us the global context without need of self-attention,giving us best of both the worlds!(beats CNN in global context,Transformers in computational effeciency)

Novelty in Approach

Mamba+UNet models has been used in 2026 for tasks like lung nodule segmentation on LIDC-IDRI data, as seen in recent papers.

Our Simpler yet **Novel** Approach !!

We use a method:

- Add noise to CT scan images.
- Gradually remove noise through diffusion steps.
- Let pure Mamba guide the cleanup for clear nodule shapes.
- Get the final segmentation mask.

Why It's Better

No need for Unet as this approach is lighter, handles tricky nodule shapes via diffusion model, and memory efficient.